Hospital ERP SaaS — Software Requirements Specification (SRS)

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This SRS specifies the observable behavior of the whole hospital ERP SaaS: 292 FR/NFR requirement IDs, each with a status, verification method and source. This tab holds the shared rules every module obeys; each module has its own tab under Modules, and non-functional requirements have their own tab.

Draft for review, version 0.4.0. Owner: technical architect + domain lead. Thin areas (finance, payroll, OT, radiology) are written as explicit gap statements, not invented rules. A requirement marked OPEN must not be built beyond its stated interim handling.

1. Scope and structure

The SRS covers the whole product (C-001), with detail concentrated where sources and decisions support it: patients, scheduling, billing, lab, beds and nursing. Thin areas such as finance, payroll, OT and radiology are written as **explicit gap statements** rather than invented rules.

Each module tab contains, where the sources allow:

1. Purpose and scope
2. Actors
3. Requirement table
4. Use cases with preconditions, inputs, outputs, state transitions and normal / alternative / denied / cancellation / recovery paths
5. State machines
6. Permissions and approvals
7. Fixed rules vs configurable policies
8. Given/When/Then acceptance examples
9. Traceability
10. Gaps and open questions

2. Requirement conventions

Every requirement has a stable ID, a status, a verification method and a source.

- **ID:** `FR-<MODULE>-<nnn>`. v0.3 IDs (e.g., FR-BIL-001) keep their meaning. New IDs continue the numbering. IDs are never reused.
- **Status:** CONFIRMED / SOURCE-DERIVED / PROPOSED / OPEN / DEFERRED (vocabulary in the Decisions & Open Questions doc, §1).
  - A requirement whose *capability* is confirmed but whose *rule values* are open is marked `CONFIRMED (values OPEN: Q-nnn)`.
  - A requirement whose behavior cannot be specified is marked **OPEN**. It must not be implemented beyond the stated interim handling.
- **Wording:** "shall" marks a requirement; "should" a recommendation; "may" an option.
- **Verification method: T** automated test (Automated Test Plan H) · **U** manual/UAT (Manual/UAT Plan I) · **I** inspection or code review · **A** analysis or measurement (load test, query plan, restore drill).
- **Source:** decision IDs (C-*), source requirement IDs (SRC*-, defined in the v0.3 baseline SOURCE_REQUIREMENTS) and proposals (P-*).
- **Amounts in examples** are synthetic (for example D = 500.00 BDT). They are test fixtures, not real prices.

3. Actors

Actor	Type	Description
Receptionist	Staff	Front-desk registration, scheduling, check-in, direct diagnostic ordering.
Cashier	Staff	Records payments, prints receipts, runs a cashier session.
Billing supervisor	Staff	Approves billing corrections within configured limits (rules OPEN).
Doctor	Staff (clinician)	Consults, orders, prescribes, admits, discharges medically, signs clinical documents.
Nurse	Staff (clinician)	Ward worklist, indents, returns, permitted observations.
Lab technologist	Staff	Collects, receives, rejects specimens; enters results.
Lab authority	Staff (clinician)	Verifies, releases, amends results (mapping OPEN, Q-009).

Actor	Type	Description
Pharmacist / pharmacy staff	Staff	Dispenses, accepts returns.
Store keeper	Staff	Stock receipt, transfers, adjustments (approval separate).
Radiology / OT / emergency / Daycare staff	Staff	Module-specific actions (later modules).
Finance officer	Staff	Reconciliation, financial reports, day-close review.
HR officer	Staff	Employee records.
Hospital administrator	Staff	Users, roles, configuration, price lists, templates. Not automatically clinical.
Management viewer	Staff	Read-only dashboards and reports.
Vendor operator	Vendor	Control plane: provisioning, entitlements, service health. Patient data only via support grant.
Patient / proxy	External	Portal (later, Q-033).
System worker	System	Background jobs acting under a recorded originating actor or system identity.
External system	System	Analyzer, payment provider, SMS gateway (through adapters, C14).

4. Global preconditions for every command and query

Unless a use case says otherwise, every request shall satisfy all eight checks **on the server** (C-029).

1. **GP-1 Authenticated session:** the session is valid, not expired and not revoked (FR-ACC-001).
2. **GP-2 Active membership:** the session's selected tenant membership is active and the user is not locked (FR-ACC-002, FR-ACC-010).
3. **GP-3 Tenant scope:** every referenced record belongs to the session's tenant. A record of another tenant is reported as *not found*, with no existence leak (NFR-ISO-001).
4. **GP-4 Facility/department scope:** the record is within the membership's permitted facility and, where configured, department scope.

5. **GP-5 Permission:** the user holds the specific permission code (§8). Menu visibility is irrelevant.
6. **GP-6 Entitlement:** the tenant has the module enabled (FR-SAA-004). An entitlement is necessary but never sufficient.
7. **GP-7 State:** the target record is in a state that allows the transition, and for state-changing commands the client's expected version matches (optimistic concurrency).
8. **GP-8 Trusted values:** amounts, prices, signer identity, tenant ID and timestamps of record are computed or read on the server. Client-supplied values for these fields are ignored or rejected.

5. Fixed product rules

These fifteen rules apply in every tenant and are not configurable.

ID	Rule	Basis
FPR-01	Tenant data is never readable or writable from another tenant, through any surface: UI, API, job, file, cache, export, report, log.	C-001, C-029
FPR-02	Appointments, visits, admissions and bed occupancies are distinct records with separate lifecycles.	C-007, C-014
FPR-03	The doctor-fee invoice and the outpatient service invoice are separate invoices with independent balances. A payment allocated to one never changes the other's balance.	C-008
FPR-04	Each payment keeps the method actually used. Corrections keep the original.	C-010, SRC-B04
FPR-05	A prescription never moves hospital stock. Only dispensing does. Outside fulfilment never moves stock.	C-012
FPR-06	Posted charges, payments, allocations, stock movements, released results and signed clinical documents are never edited or deleted. Corrections are new records (reversal, adjustment, amendment) linked to the original.	C-029, P-013
FPR-07	Viewing, printing, exporting or navigating never changes a record's business state.	SRC-L12–L15

ID	Rule	Basis
FPR-08	A repeated identical command (same idempotency key and payload) returns the original outcome without repeating side effects. A different payload under the same key is rejected.	C-029
FPR-09	A source event posts at most one charge (unique source posting reference).	SRC-B06
FPR-10	A physical bed has at most one active occupancy at any instant. A bed vacated by discharge is unavailable until cleaning is recorded.	C-016, P-012
FPR-11	Money is exact decimal with explicit currency. No floating-point arithmetic. Rounding happens only where a configured rule says so.	P-011
FPR-12	Every sensitive action is audited with actor, tenant, target, version, time and outcome.	C-029
FPR-13	The software never decides clinical need, triage category, diagnosis or dose. Staff record decisions; the software records who and when.	C-015, C-027
FPR-14	No patient may be merged automatically.	Identity agreements
FPR-15	"Saved" or "success" feedback appears only after the server confirms the commit.	SRC-I05, v0.3 DESIGN

6. Configurable hospital policies

Configurable policies are **tenant configuration with effective dates and versions** (P-004, FR-CFG-002). A record keeps the policy version in force when it was created. A policy whose *values* are OPEN ships with the safest interim value listed here, labelled PROPOSED. Changing it requires a tenant administrator with `admin.config.publish`.

ID	Configurable policy	Scope	Interim / default value	Status	Owner question
CP-01	Directly orderable tests (reception may order without consultation)	Per test	No test directly orderable until configured	CONFIRMED capability (C-006)	None
CP-02	Payment eligibility rule per service	Per service / category	`FULL_PAYMENT`	CONFIRMED capability (C-011); deposit values OPEN	Q-004

ID	Configurable policy	Scope	Interim / default value	Status	Owner question
CP-03	Deposit rules (minimum, which services, allocation order)	Per service category	Disabled	OPEN	Q-004, Q-040
CP-04	Emergency payment exceptions	Per facility	Disabled	OPEN	Q-005
CP-05	Price lists (services, tests, doctor fees)	Per tenant, effective-dated	Must be configured before go-live	CONFIRMED capability	Q-038
CP-06	Discount reasons, limits, approvers	Per role	Discounts disabled	OPEN	Q-019
CP-07	Rounding rule for invoices/receipts	Per tenant	None (exact 2-decimal)	OPEN	Q-020
CP-08	Refund approval levels and methods	Per tenant	Refunds disabled	OPEN	Q-021
CP-09	MRN / invoice / receipt / specimen numbering patterns	Per document type	`{PREFIX}-{YYYY}-{seq:6}` (synthetic)	CONFIRMED capability; format OPEN	Q-034
CP-10	Required registration fields (above product minimum)	Per tenant	Product minimum only (FR-PAT-005)	OPEN	Q-035
CP-11	Slot capacity, overbooking, queue numbering	Per schedule	Fixed capacity, no overbooking, sequence at check-in	OPEN	Q-037
CP-12	Session timeouts, lockout threshold, MFA roles	Per tenant	Idle 30 min, absolute 12 h, lockout after 5 failures for 15 min, MFA for admin/finance roles (all PROPOSED)	OPEN	Q-051
CP-13	Lab verify/release separation (same person allowed?)	Per test category	Same person **not** allowed	OPEN	Q-009
CP-14	Lab reference ranges, methods, templates, units	Per test version	Entered by lab; no product defaults	CONFIRMED capability (FR-LAB-003)	Q-011
CP-15	Bed cleaning required after transfer vacancy	Per tenant	Required	OPEN	Q-008
CP-16	Bed-rent rule	Per tenant	No automatic charge; manual posting	OPEN	Q-007
CP-17	Report delivery allowed with outstanding dues	Per report type	Allowed, dues shown	OPEN	Q-013

ID	Configurable policy	Scope	Interim / default value	Status	Owner question
CP-18	UI default language; print language per template; digits; date format	Per tenant / user / template	User choice; Western digits; `dd-MM-yyyy`	CONFIRMED capability (C-025); values OPEN	Q-050
CP-19	Print templates (A4/A5/label sizes, header, logo)	Per document type	Product templates	CONFIRMED capability (SRC-Q01)	Q-046
CP-20	Pharmacy OPD payment gate / IPD posting	Per tenant	OPD pay-before-issue; IPD to running account (disabled until confirmed)	OPEN	Q-015
CP-21	Medicine return acceptance before credit	Per tenant	Credit after acceptance	OPEN	Q-018
CP-22	Order cancellation allowed stages and approvers	Per service category	Before payment and before work only	OPEN	Q-017
CP-23	Emergency priority category list	Per facility	Tenant-defined list, no product defaults	OPEN	Q-025
CP-24	Segregation-of-duty pairs (e.g., issuer ≠ collector)	Per tenant	None enforced except CP-13	OPEN	Q-039

7. Standard paths for every state-changing command

Each use case lists only its specific paths and refers to these twelve by ID.

ID	Situation	Required behavior	Error code
SP-01	**Duplicate submission** (double-click, retry after timeout) with the same idempotency key and identical payload	Return the original result. No new side effect.	None (original response, `Idempotent-Replay: true` header)
SP-02	Same idempotency key, **different payload**	Reject. Nothing changes.	`IDEMPOTENCY_KEY_REUSED`

ID	Situation	Required behavior	Error code
SP-03	**Stale version** (another user changed the record)	Reject. The UI reloads and shows what changed.	`STALE_RECORD`
SP-04	**Permission denied**	Reject. Audit sensitive denials (FR-AUD-002). The UI shows "You do not have permission to …" without revealing data.	`PERMISSION_DENIED`
SP-05	**Out of scope** (other tenant, other facility, non-existent)	Reject as not found.	`NOT_FOUND`
SP-06	**Invalid transition** (state does not allow it)	Reject with current state.	`INVALID_TRANSITION`
SP-07	**Validation failure** (missing or invalid field)	Reject with field-level messages in the user's language.	`VALIDATION_FAILED`
SP-08	**Uncertain outcome** (network timeout; client does not know if the server committed)	The client re-sends with the **same** idempotency key, or queries status by key. It never creates a new key for the same intent.	None
SP-09	**Server/database unavailable**	Nothing is committed (transaction rolled back). The UI shows "Not saved — try again" and keeps the form data.	`SERVICE_UNAVAILABLE`
SP-10	**Concurrent conflicting command** (two cashiers, two bed allocations)	Exactly one succeeds. The other gets SP-03 or a domain-specific code (`BED_UNAVAILABLE`, `AMOUNT_EXCEEDS_BALANCE`).	As stated

ID	Situation	Required behavior	Error code
SP-11	**Session expired mid-form**	The form is kept locally (no business data in persistent browser storage). After re-login the user resubmits with the same idempotency key.	`UNAUTHENTICATED`
SP-12	**Cancellation by user before submit**	Nothing is sent. Unsaved-changes warning on navigation.	None

8. Permission catalogue and approval responsibilities

The permission *codes* are PROPOSED; the *separation* of these rights is CONFIRMED (C-029, v0.3 ACCESS_CONTROL); *which role* receives each code is OPEN (Q-003).

Default template abbreviations (PROPOSED role templates): RCP Receptionist · CSH Cashier · BSV Billing supervisor · DOC Doctor · NUR Nurse · LBT Lab technologist · LBA Lab authority · PHA Pharmacy · STK Store keeper · FIN Finance · ADM Hospital administrator · MGT Management viewer. "Approval" marks actions that are themselves approvals or need an approver.

Permission code	Action	Default template	Approval	FR
`patient.record.search`	Search patients	RCP, CSH, DOC, NUR, LBT, LBA, PHA	No	FR-PAT-001
`patient.record.register`	Register a patient	RCP	No	FR-PAT-002
`patient.identity.edit`	Edit identity fields	RCP (limited fields), ADM	Reason required	FR-PAT-008
`patient.identity.reconcile`	Link a temporary identity to a known patient	ADM (proposed)	Yes (Q-036)	FR-PAT-004
`patient.sensitive.view`	View unmasked NID	ADM, RCP (proposed)	Audited read	FR-PAT-007
`practitioner.manage`	Create/edit practitioners	ADM	No	FR-PRA-001
`scheduling.schedule.manage`	Manage schedules/sessions	ADM, RCP (proposed)	No	FR-SCH-002

Permission code	Action	Default template	Approval	FR
`scheduling.appointment.book`	Book/reschedule/cancel	RCP	No	FR-SCH-002
`visit.checkin`	Check in an appointment	RCP	No	FR-VIS-002
`visit.create`	Create walk-in / diagnostic visit	RCP	No	FR-VIS-001
`order.direct_diagnostic.create`	Reception direct test orders	RCP	Per-test policy CP-01	FR-ORD-001
`order.service.create`	Clinician orders	DOC	No	FR-ORD-002
`order.service.cancel`	Cancel an order	DOC, RCP (before payment)	Stage rules (Q-017)	FR-ORD-006
`clinical.consultation.record`	Write consultation note	DOC	No	FR-CLI-003
`clinical.document.sign`	Sign clinical document	DOC	No	FR-CLI-003
`clinical.document.amend`	Amend signed document	DOC (author)	Reason required	FR-CLI-003
`clinical.prescription.create`	Create/sign prescription	DOC	No	FR-CLI-004
`billing.invoice.issue`	Issue invoices / add lines	RCP, CSH (Q-039)	No	FR-BIL-007
`billing.payment.record`	Record payment and print receipt	CSH	No	FR-BIL-011
`billing.payment.method_correct`	Correct a payment method	BSV	Yes, reason	FR-BIL-025
`billing.discount.apply` / `billing.discount.approve`	Discounts	BSV	Yes (Q-019)	FR-BIL-013/014
`billing.refund.request` / `billing.refund.approve`	Refunds	CSH / BSV	Yes (Q-021)	FR-BIL-018
`billing.line.cancel`	Cancel/reverse an invoice line	BSV	Yes (Q-017)	FR-BIL-020
`billing.eligibility.override`	Emergency exception	None (disabled)	Yes (Q-005)	FR-BIL-016

Permission code	Action	Default template	Approval	FR
`billing.cashier_session.manage`	Open/close own session	CSH	No	FR-BIL-026
`billing.cashier_session.review`	Review/accept variances	FIN	Yes	FR-BIL-026
`lab.specimen.collect` / `lab.specimen.receive` / `lab.specimen.reject`	Specimen handling	LBT	No	FR-LAB-004..007
`lab.result.enter`	Enter results	LBT	No	FR-LAB-008
`lab.result.verify`	Verify	LBA	Approval (Q-009)	FR-LAB-009
`lab.result.release`	Release	LBA	Approval (Q-009)	FR-LAB-010
`lab.result.amend`	Amend released result	LBA	Reason; approval (Q-011)	FR-LAB-012
`lab.config.manage`	Tests, ranges, templates	LBA, ADM	Publish is versioned	FR-LAB-016
`lab.report.deliver`	Record report delivery	RCP (delivery desk), LBT	No	FR-LAB-022
`ipd.admission.create`	Create admission	RCP, DOC (admission request)	No	FR-IPD-001
`ipd.bed.allocate` / `ipd.bed.transfer`	Allocate / transfer	NUR (ward in-charge), RCP (admission desk)	No	FR-IPD-002, FR-IPD-007
`ipd.bed.cleaning.complete`	Mark cleaning complete	NUR / housekeeping	No	FR-IPD-003
`ipd.discharge.medical`	Medical discharge	DOC	No	FR-IPD-011
`ipd.discharge.financial_clear`	Financial clearance	CSH / BSV	Policy Q-006	FR-IPD-012
`ipd.discharge.depart`	Record departure	NUR, RCP	No	FR-IPD-013
`nursing.worklist.view`	Ward worklist	NUR	No	FR-NUR-001

Permission code	Action	Default template	Approval	FR
`nursing.indent.manage`	Create/edit indents	NUR	No	FR-NUR-004
`nursing.return.submit`	Submit medicine return	NUR	No	FR-NUR-005
`pharmacy.dispense`	Dispense	PHA	No	FR-PHA-004
`pharmacy.return.accept`	Accept return	PHA	Approval of quantity	FR-PHA-007
`inventory.stock.adjust.request` / `.approve`	Stock adjustment	STK / store manager	Yes	FR-INV-004
`inventory.transfer.dispatch` / `.receive`	Transfers	STK	No	FR-INV-005
`inventory.grn.receive`	Goods receipt against PO	STK	No	FR-INV-009
`report.view.<area>`	View reports by area	Varies	No	FR-REP-004
`report.export`	Export to file	FIN, ADM, MGT	Audited	FR-REP-004
`admin.user.manage`	Users and memberships	ADM	No	FR-ACC-008
`admin.role.manage`	Roles and permissions	ADM	Audited	FR-ACC-008
`admin.config.publish`	Publish configuration/policy versions	ADM	Audited; effective date	FR-CFG-002
`admin.price.publish`	Publish price list versions	ADM / FIN	Audited	FR-CFG-002
`audit.view`	View tenant audit log	ADM	No	FR-AUD-005
`support.grant.approve`	Approve a vendor support grant	ADM	Yes	FR-SUP-001
`vendor.tenant.provision` (control plane)	Provision tenants	Vendor operator	No	FR-SAA-002

Approval responsibilities

- An approval is recorded as its own event, with approver, time, reason and the approved record's version.
- An approver cannot approve their own request when a segregation rule (CP-13, CP-24) applies.
- **Approval rules whose values are OPEN are disabled, not guessed.** For example, discount limits stay disabled until Q-019 is answered.

9. Error codes, use-case format and traceability

Error codes are stable strings in `packages/contracts`, mapped to HTTP status in the Architecture doc (API contracts, §8.3). Messages are localized (bn/en).

`UNAUTHENTICATED`, `PERMISSION_DENIED`, `NOT_FOUND`, `VALIDATION_FAILED`, `INVALID_TRANSITION`, `STALE_RECORD`, `IDEMPOTENCY_KEY_REUSED`, `PAYMENT_NOT_ELIGIBLE`, `AMOUNT_EXCEEDS_BALANCE`, `BED_UNAVAILABLE`, `BED_NEEDS_CLEANING`, `CAPACITY_FULL`, `INSUFFICIENT_STOCK`, `BATCH_EXPIRED`, `ENTITLEMENT_MISSING`, `POLICY_NOT_CONFIGURED`, `SEGREGATION_OF_DUTY`, `ACCOUNT_LOCKED`, `SERVICE_UNAVAILABLE`, `FEATURE_NOT_AVAILABLE`.

`FEATURE_NOT_AVAILABLE` is deliberately visible. When a user tries an action whose rule is OPEN (for example a refund before Q-021), the UI says the hospital has not configured it. It never offers a guessed workflow.

Use-case format used in every module tab:

```
UC-<MOD>-nn  Title
Actors / permission:
Preconditions:          (in addition to GP-1..GP-8)
Trigger:
Inputs:                 (client-supplied; trusted values computed on server)
Normal path:            numbered steps
Outputs:                records created/changed, events, prints
State transitions:
Alternative paths:      A1, A2 …
Denied paths:           D1 … (plus SP-04/SP-05)
Cancellation:           C1 …
Recovery:               R1 … (plus SP-01/SP-08/SP-09)
Requirements:           FR-…
```

Traceability. Each module tab ends with a traceability table: requirement → decision/source → API command → data entity → permission → test IDs. The consolidated objective-to-gate chains and readiness state are in the Traceability & Readiness document (N) of the full pack.